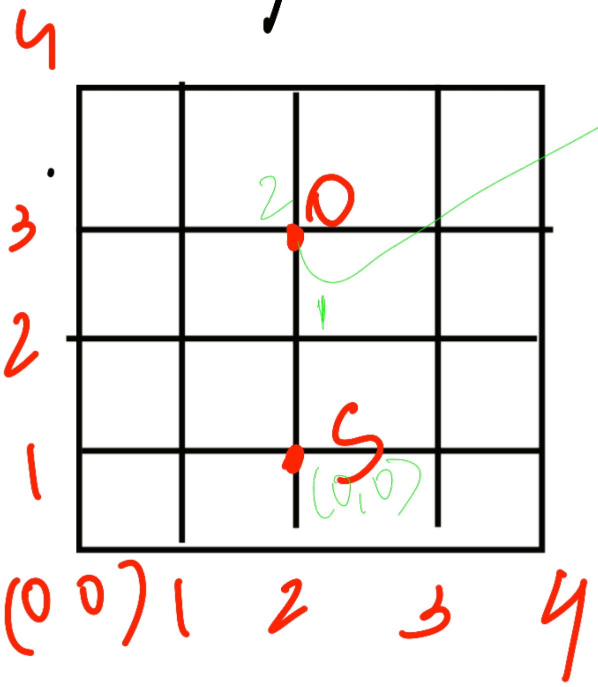
* Global / local co-ordinate system



(0,2)

global co-ordinates

Sensor → (2,1) (0,0)

Object wrt to sensor ⇒ (0,2)

global co-ordinates (object) → + O_c wrt S

⇒ (2,1) + (0,2)

⇒ (2,3)

(Basic)

* Rotation Matrix

→ Sensor when not aligned to global axes → usage of this

assume rotated by θ

$$R(\theta) = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$$

$$P_{rotated} = R * local.$$

$$\therefore P_{global} = P_{sensor} + R(\theta) * local$$